

Fundamentals of Cyber Defense & SOC Operations

Phase 1

NAME : VISHWAS H N , SRN : PES1UG23CS701 , SECTION : L

NAME : SHESHAGIRI S , SRN : PES1UG24CS832 , SECTION : H

Organization Chosen

Automated High-Frequency Trading (HFT) Firm Operating an Electronic Stock Trading Floor with an Integrated SOC

1. Organization Overview

The selected organization is an **Automated High-Frequency Trading (HFT) Firm** that operates an **electronic stock trading floor**. The firm uses **algorithm-driven trading systems** to execute a large number of buy and sell orders at extremely high speeds, often within milliseconds.

The trading floor is fully automated and consists of **trading algorithms, market data feeds, order execution engines, and risk management systems** hosted on low-latency servers. Due to the speed and volume of trading, even minor system anomalies or security breaches can cause **market instability**, including events similar to **flash crashes**.

To ensure trading integrity, regulatory compliance, and system reliability, the organization maintains a **Security Operations Center (SOC)** that continuously monitors logs, detects abnormal trading behavior, and responds to incidents using **SIEM and SOAR-based workflows**.

2. Asset Identification & CIA Mapping

Asset List :

- Trading Algorithms
- Market Data Feed Systems
- Order Execution Engine

- Risk Management & Circuit Breaker System
- Trading Servers
- Trader Monitoring Workstations
- Network Infrastructure
- Trade and Audit Logs
- SOC Monitoring Systems

CIA MAPPING

Asset	Confidentiality	Integrity	Availability
Trading Algorithms	High	Very High	High
Market Data Feeds	Medium	Very High	Very High
Order Execution Engine	Medium	Critical	Critical
Risk Management System	Medium	Critical	High
Trading Servers	Medium	High	Critical
Trader Workstations	High	High	Medium
Network Infrastructure	Low	High	Critical
Trade Logs	High	High	Medium
SOC Systems	Medium	High	High

Justification:

Integrity and availability are the most critical security properties, as incorrect or delayed trades can cause financial loss and market instability.

3. Threat Actor Identification

Threat Actor Table

Threat Actor	Motivation	Capability
Malicious External Attacker	Financial gain	Medium
Rogue Insider (Trader/Developer)	Profit manipulation	High
Competing HFT Firms	Market advantage	High

Cybercriminal Groups	Financial theft	Medium
Nation-State Actors	Market disruption	Advanced

4. Risk Identification

Risk 1: Flash Crash Triggered by Algorithm Manipulation

A compromised or faulty trading algorithm can initiate massive sell or buy orders, causing a sudden collapse in prices.

Impact : Market instability, regulatory scrutiny, financial loss.

Risk 2: Market Data Feed Tampering

Attackers may inject false price data, causing algorithms to make incorrect trading decisions.

Impact: Incorrect trades, cascading algorithm failures.

Risk 3: Insider Abuse of Privileged Access

Developers or traders may intentionally alter algorithm parameters for personal gain.

Impact: Market manipulation, legal consequences.

Risk 4: Denial-of-Service on Trading Infrastructure

Network or application-level attacks may delay or block trade execution.

Impact: Loss of availability and missed trading opportunities.

Risk 5: Log Tampering and Evidence Removal

Attackers may delete or modify logs to hide manipulation.

Impact: Delayed detection and failed compliance audits.

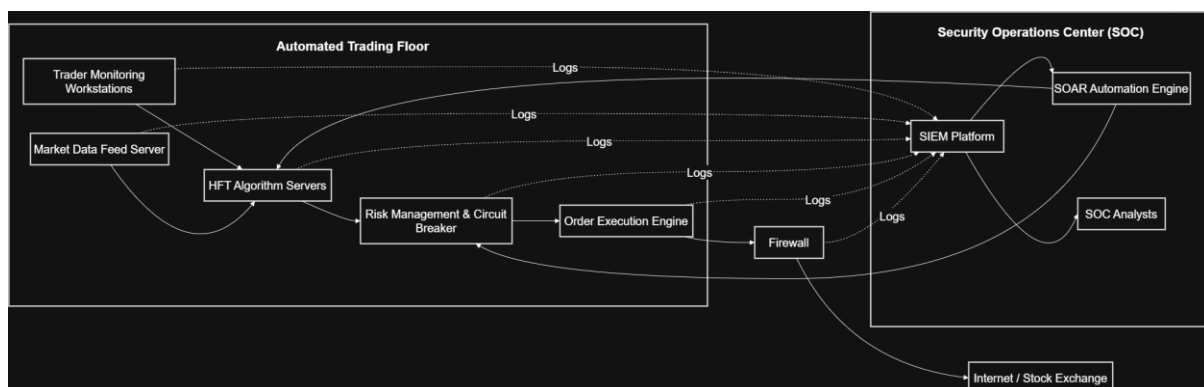
5. Protocol–Threat Mapping

Protocol	Associated Threat
HTTP/HTTPS	API abuse, unauthorized order placement
TCP	Session hijacking, latency manipulation
DNS	Market feed redirection
NTP	Time desynchronization affecting trade order
SSH	Unauthorized server access

6. Basic Network Architecture

Components included in the network:

- Trader Monitoring Workstations
- Algorithm Trading Servers
- Market Data Feed Server
- Risk Management Server
- Firewall
- Internet / Stock Exchange Gateway
- SOC / SIEM Server



Description:

All trading and infrastructure components forward logs to the SIEM for centralized monitoring by the SOC.

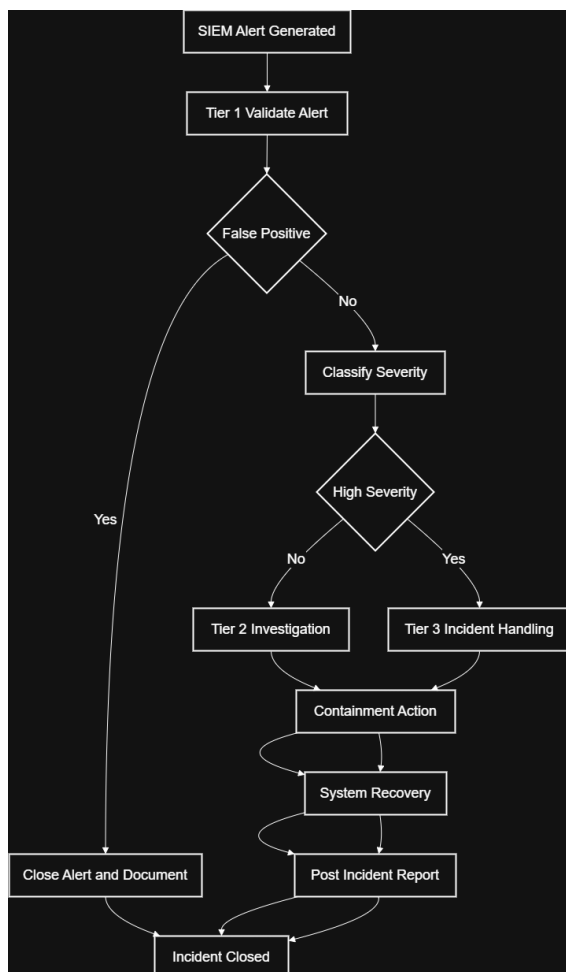
7. Log Source Identification

Log Source Table

Log Source	Log Type	Purpose
Trading Servers	Linux Syslogs	Algorithm behavior analysis
Order Execution Engine	Application Logs	Trade validation
Risk System	Audit Logs	Circuit breaker monitoring
Trader Workstations	Windows Security Logs	Insider threat detection
Firewall & Router	Network Logs	Traffic anomaly detection
SIEM	Correlated Alerts	Incident response

8. SOC Structure & Workflow

SOC Workflow



SOC Tier Roles

Tier 1 SOC Analyst

The Tier 1 SOC Analyst validates SIEM-generated alerts and performs initial triage to identify false positives. Non-incidents are documented and closed to reduce alert fatigue and ensure efficient SOC operations.

Tier 2 SOC Analyst

The Tier 2 SOC Analyst conducts detailed investigation of confirmed alerts by correlating logs and events across multiple systems. This role supports containment actions for non-critical incidents before escalation if required.

Tier 3 SOC Analyst

The Tier 3 SOC Analyst handles high-severity incidents by leading incident response and containment activities. This role also performs forensic analysis, root-cause identification, and oversees system recovery and security tuning.

9. Role–Responsibility Matrix

Role	Responsibility
Tier 1 SOC Analyst	Alert validation, false positive identification, initial triage
Tier 2 SOC Analyst	Detailed investigation, log correlation, incident analysis
Tier 3 SOC Analyst	High-severity incident response, forensics, recovery oversight
SOC Engineer	SIEM and SOAR rule tuning, platform maintenance
Trading Operations Team	Business impact validation and trading decision support
Compliance Officer	Regulatory reporting and audit coordination

10. SOC Metrics Definition

Mean Time to Detect (MTTD)

MTTD measures the average time taken by the SOC to detect abnormal trading activity or potential manipulation.

Importance : Lower MTTD helps prevent flash-crash-like events.

Mean Time to Respond (MTTR)

MTTR measures the time taken to contain and resolve a detected incident.

Importance : Lower MTTR minimizes financial loss and market impact.

Why are MTTD and MTTR important?

In an **Automated High-Frequency Trading (HFT) environment**, security incidents and system anomalies can cause severe financial and market impact within **seconds**. Therefore, measuring how quickly incidents are detected and resolved is critical.

Importance of MTTD (Mean Time to Detect)

MTTD is important because it determines how fast the SOC can **identify abnormal trading behavior**, such as algorithm malfunction, market data manipulation, or insider misuse. A lower MTTD allows the organization to detect early warning signs before they escalate into **flash-crash-like events**, preventing large-scale financial losses and market instability.

Importance of MTTR (Mean Time to Respond)

MTTR is important because it measures how quickly the SOC can **contain and resolve an incident** after detection. In an HFT firm, faster response times help immediately disable rogue algorithms, trigger circuit breakers, and restore safe trading operations, thereby minimizing financial damage, regulatory violations, and reputational harm.

Overall Importance

Together, MTTD and MTTR reflect the **effectiveness and maturity of the SOC**. In a high-speed trading environment where milliseconds matter, low MTTD and MTTR ensure **system integrity, trading continuity, and regulatory compliance**, making them critical metrics for protecting against flash crash manipulation.

11. Conclusion

This project demonstrates how **cyber defense principles and SOC operations** can be applied to an **automated high-frequency trading environment**. By integrating SIEM and SOAR, the organization ensures **algorithm integrity, market stability, and regulatory compliance**, while protecting against flash crash manipulation and cyber threats.